

Amortized Bayesian parameter recovery for dynamical systems with conditional flow matching

The `amortix` package: architecture, evaluation methodology, and benchmarks

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Provenance of the measurements. All numbers in this report come from package commit `f36e4d`. That version samples on a uniform time grid with 20 midpoint steps, and it puts the observed value into the token as it is (“raw”). Two later changes keep these numbers and change the package. Commit `d35f9c4` makes the bare-point embedding the default and the consecutive-pair embedding an explicit option (`embed="wfilm"`). The sampler and the token value coordinate have no default values any more, and the sampler has a second time grid (Sec. 4).

Abstract

This report presents a package for amortized Bayesian estimation of the parameters of dynamical systems. The systems are ODEs, SDEs and discretized PDEs. The data are noisy observations at a finite set of time points. The package trains a conditional flow-matching model once for each system, on simulated pairs. The trained model returns posterior samples for any observation set, with any number and any placement of points, in about 70 milliseconds. We test the package on fourteen systems. Each system has an exact or a validated reference posterior. Twelve of the fourteen systems match their reference to a median Fréchet distance of 0.003–0.08 in reference-normalized coordinates. The tools that build and validate the reference posteriors are part of the package.

1 Introduction

A modeler writes a dynamical system with unknown physical constants. The system is an ODE, an SDE or a PDE. The modeler observes one trajectory at a finite set of time points and asks what the observations say about the constants. This is one of the basic inverse problems of applied science. In practice the problem is ill-posed. Different parameter values can give almost the same observations. This happens most often when the state is partially observed or the sampling is sparse. A point estimate then hides the ambiguity and is unstable under noise. The Bayesian formulation reports the full posterior distribution over the parameters, given the data. Ambiguity becomes posterior spread and correlation. The price is two computational obstacles. First, for most stochastic and partially observed systems the likelihood of the observations is intractable or expensive, because it integrates over unobserved trajectory segments. Second, modern workflows need posteriors many times. Calibration studies process thousands of synthetic observation sets. Sequential experiments update the posterior after each new measurement. Bayesian experimental design compares candidate observation schedules by the posteriors they give. In such loops, MCMC or optimization for each instance is the dominant cost, even when a likelihood is available.

Simulation-based inference (SBI) solves both problems with one idea (Cranmer et al., 2020). It learns a conditional generative model of the parameters given the data from simulated pairs. The parameters come from the prior and the observation sets come from the simulator. The cost of inference moves into one training phase. Neural posterior estimation (NPE) and its sequential variants fit normalizing flows by maximum likelihood (Papamakarios and Murray, 2016; Greenberg et al., 2019). Flow-matching and diffusion variants use transport-based generative models instead (Wildberger et al., 2023; Gloeckler et al., 2024). Mature software exists: the `sbi` toolkit (Boelts et al., 2024), BayesFlow (Radev et al., 2020) and the `sbi` benchmark collection (Lueckmann et al., 2021).

These toolkits leave three problems open for dynamical systems. The three problems are connected. First, the conditional density estimators consume a fixed-length data vector after global standardization. The estimator sees the coordinates the user chose. On multiplicative and heavy-tailed processes (prices, concentrations, populations) this convention fails in reproducible ways. With prescribed coordinates the estimator becomes biased or mis-calibrated (Sec. 3.3). Second, a fixed-length interface cannot give the

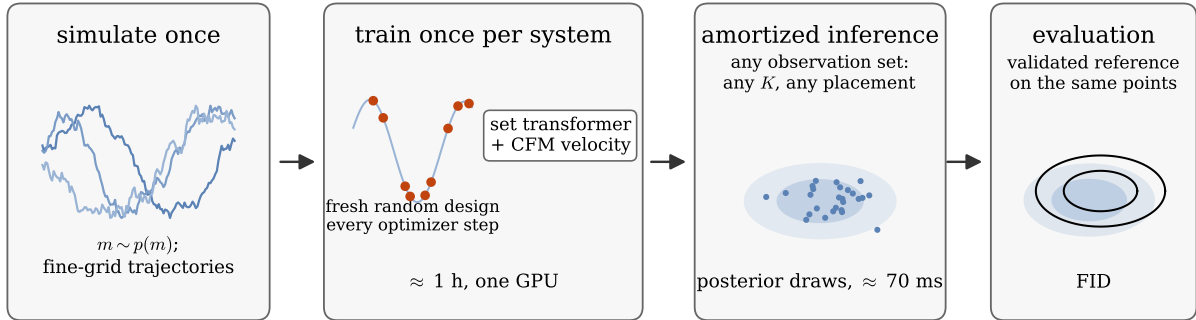


Figure 1: The approach. Parameters come from the prior and are simulated once on a fine grid (left). Training fits a conditional flow-matching posterior with a set-transformer encoder. Each optimizer step draws a fresh random observation design (Secs. 3, 4). The trained network returns posterior samples for any observation set of the experiment, with any number and any placement of points. We measure the accuracy against exact or validated reference posteriors computed on the same observed points (Sec. 5).

object that sequential refinement and Bayesian experimental design need. That object is one posterior conditioned on an arbitrary subset of the measurement points. Third, the standard evaluation practice is simulation-based calibration (Talts et al., 2018). This is a test of a necessary condition. Its statistical power at practical sample sizes is limited. A learned posterior can be confidently wrong in ways that rank diagnostics do not detect (Hermans et al., 2022). A failure of the first kind can pass the accepted diagnostics.

This report describes **amortix**, a compact PyTorch package. It has about 3,500 lines and one default model of 4.8×10^5 weights. Three decisions answer the three problems, in order. The input representation is learned as part of the model. One network is trained over random observation designs. Accuracy is measured against reference posteriors, not against self-consistency. The main contributions are:

- *A learnable input representation* (Sec. 3). It has three parts: a monotone mixture-of-scales reparametrization of the value axis, attention phases computed from the physical observation times, and explicit design-size features. A measured failure in the conditioning of the encoder input motivates each part. Each part is a learnable repair, documented with the part itself (Sec. 3.3).
- *Design amortization* (Secs. 4, 7.3). One network approximates $p(m \mid S)$ for every subset S of the measurement points. The subsets have arbitrary sizes and placements, and they include spatio-temporal sensor designs for PDEs. The training procedure resamples the designs at every optimizer step, under a measured design-size law.
- *An evaluation methodology with references as the record* (Sec. 5). All fourteen benchmark systems have exact or validated reference posteriors. The engine that the likelihood admits builds each reference: closed forms, adaptive Metropolis, nested sampling, or tempered SMC over particle-filter estimates. Three independent checks admit each reference. Simulation-based calibration is only a screen. The tools ship with the package.
- *Measurements* (Sec. 7). We report the accuracy on the suite and its dependence on model size, simulation budget and data reuse. We report the full cost (Table 3). Training takes 58–82 minutes per system on one GPU. A laptop CPU gives bit-identical results at four times the wall time. After training, one posterior costs about 70 ms. The per-instance reference samplers cost milliseconds to one hour per observation set. For a single dataset with a cheap likelihood, classical sampling stays the right tool. Amortization pays when posteriors repeat or when the likelihood is intractable.

Appendix D maps the terms of this report to package options.

2 Method

Figure 1 shows the approach. This section defines each element.

2.1 Problem formulation

A dynamical system depends on a parameter vector $m \in \mathbb{R}^d$ with a known prior $p(m)$. In all benchmarks the prior is uniform on a box. The system gives a state trajectory $x(t)$. The trajectory comes from an ODE $\dot{x} = f(x; m)$, from an SDE $dx = f(x; m)dt + \sigma(x; m)dW_t$, or from a spatially discretized PDE. A noisy channel observes the trajectory at K points:

$$y_i = h(x(t_i)) + \varepsilon_i, \quad \mathcal{D} = \{(t_i, y_i)\}_{i=1}^K. \quad (1)$$

We call $\mathcal{D}$ the *observation set* of one experiment. It is the entire input to inference. Training and evaluation use many independently simulated observation sets. Each set is paired with the parameter that generated it. The observation operator h can expose only part of the state. In the benchmarks below, h gives only the price of an asset, and the stochastic volatility stays latent. Or it gives only the membrane voltage of a neuron, and the gating variables stay latent. Or it gives only three spatial sensors of a PDE field. The target is the posterior $p(m | \mathcal{D}) \propto p(\mathcal{D} | m)p(m)$.

Three assumptions define the setting. The parameter is finite- and low-dimensional ($d = 2$ –5 in every benchmark). The prior factorizes across components and has tractable marginal CDFs (uniform boxes here; the construction extends to any such prior). The simulator runs forward cheaply and repeatedly, and nothing else is required of it: no likelihood evaluations and no gradients. The observation times and sensor indices are known exactly. The measurement noise ε_i is independent across observations with a known per-system law.

The posterior is intractable, but sampling from the *joint* distribution of parameters and data is easy. Draw $m_j \sim p(m)$, simulate the system, apply the observation channel. This gives as many training pairs $(m_j, \mathcal{D}_j)$ as needed. A conditional generative model trained on these pairs approximates the posterior for every observation set at the same time. The cost of inference moves into one training phase. Inference for a new observation set is one forward pass. We add one requirement to standard amortization. A single trained network must stay correct for any number K and any placement of observation times (and, for PDEs, sensor positions). In formal terms, for every subset S of the measurement points of one experiment the network must approximate $p(m | S)$. The family of answers must be coherent; in particular the posterior must widen when points are removed. We call this requirement *design amortization*. Sequential refinement and Bayesian experimental design consume this object, because they rank candidate observation schedules by the posteriors the schedules give.

2.2 Posterior sampling by conditional flow matching

The method represents the posterior as a *transport*, not as a density formula. The transport is a learned, data-conditioned deformation that carries a fixed reference distribution onto the posterior. Sampling draws from the reference and applies the deformation. There is no normalizing constant to compute and no invertibility constraint in the network. The flow of a velocity field defines the deformation implicitly. Ordinary least-squares regression on simulated pairs fits the field. This construction is conditional flow matching (Lipman et al., 2023).

First, the parameters are reparametrized component by component,

$$m \mapsto \Phi^{-1}(F_{\text{prior}}(m)) \in \mathbb{R}^d, \quad (2)$$

where F_{prior} collects the marginal prior CDFs and Φ is the standard normal CDF. In the new coordinates the prior is exactly $\mathcal{N}(0, I)$ and the box constraints disappear. The uninformative limit then costs nothing: when the data carry no information, the posterior is the reference and the transport is the identity. From here on we work in these coordinates and keep the symbol m for the parameter in them. The inverse of (2) is applied exactly once, at the end of sampling, to return the draws on the original scale.

A *virtual time* $\tau \in [0, 1]$ indexes the transport. This is an algorithmic variable. It has no relation to the physical time t at which the system is observed. The symbol t appears only inside observation sets, and τ only in the transport. Subscripts in τ give the position along the transport path. The endpoint m_1 is a parameter drawn together with its observation set $\mathcal{D}$. The start $m_0 \sim \mathcal{N}(0, I)$ is pure reference noise. It is not the transform of any parameter. It is the raw material that the transport turns into a posterior draw at $\tau = 1$. Between them lies the training interpolation $m_\tau = (1 - \tau)m_0 + \tau m_1$. The velocity network learns the direction of this path by regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{m, \mathcal{D}, m_0 \sim \mathcal{N}(0, I), \tau \sim U[0, 1]} \|v_\theta(m_\tau, \tau, \mathcal{D}) - (m_1 - m_0)\|^2. \quad (3)$$

The minimizer of (3) is the marginal velocity field. Its flow transports $\mathcal{N}(0, I)$ at $\tau = 0$ to $p(m_1 | \mathcal{D})$ at $\tau = 1$ (Lipman et al., 2023). Sampling draws $m_0 \sim \mathcal{N}(0, I)$, integrates $dm/d\tau = v_\theta(m, \tau, \mathcal{D})$ from $\tau = 0$ to 1, and maps the endpoint back through the inverse of (2). The numerical settings of the integration are not details. Sec. 4 states them, and it reports a case where the usual grid fails. The same estimator family, with a different conditioning architecture, is available in `sbi` as flow-matching posterior estimation (FMPE) (Wildberger et al., 2023). This package differs in the input representation, not in the generative engine.

No stage needs likelihood evaluations, simulator gradients or per-instance optimization. The simulator runs only forward, during the generation of the training set.

The training objective is consistent in the following sense.

Proposition 1 (consistency). *Let $(m, \mathcal{D}) \sim p(m)p(\mathcal{D} | m)$ and let m_1 be given by (2). Minimize the loss (3) over all measurable velocity fields. Then for $p(\mathcal{D})$ -almost every $\mathcal{D}$ the flow of the minimizer transports $\mathcal{N}(0, I)$ at $\tau = 0$ to the conditional law $p(m_1 | \mathcal{D})$ at $\tau = 1$. Equivalently, the sampling procedure above draws exactly from the posterior (Lipman et al., 2023; Wildberger et al., 2023).*

The proposition is an idealization. A finite network, a finite simulation budget and a numerical solver remove every calibration guarantee. The residual error becomes an empirical quantity. Section 5 builds the instruments that measure it.

3 Conditioning architecture

The velocity field $v_\theta(m, \tau, \mathcal{D})$ is conditioned on an observation set of time-stamped measurements. Three properties of that set drive the architecture. The number of observations K varies from experiment to experiment, by two orders of magnitude in the variable-design setting. The observation times are arbitrary real numbers, in general not equally spaced, so positional schemes for uniform sequences do not apply. The times carry all ordering information, so the output must not depend on the storage order of the observations. These properties rule out the fixed-length-vector conditioners that NPE implementations use. A transformer that operates on one feature vector per observation meets them. We call that feature vector a *token*.

3.1 Overall scheme

The network evaluates $v_\theta(m, \tau, \mathcal{D})$. Here m is the normalized parameter vector partway along its transport path: the interpolation m_τ at training time, the current ODE state at sampling time. The scalar τ is the virtual time of the transport. The design follows the encoder–decoder pattern of conditional flow-matching and diffusion models. The data are encoded once. A decoder combines that encoding with the object being generated,

$$v_\theta(m, \tau, \mathcal{D}) = \text{Dec}(m, \tau, \text{Enc}(\mathcal{D})). \quad (4)$$

The types are explicit. `Enc` maps the observation set, the K time-stamped measurements of one experiment, to a memory of K vectors, $\text{Enc}(\mathcal{D}) \in \mathbb{R}^{K \times d_{\text{model}}}$. The memory is computed once per observation set. `Dec` : $\mathbb{R}^d \times [0, 1] \times \mathbb{R}^{K \times d_{\text{model}}} \rightarrow \mathbb{R}^d$ combines the current transport point and the virtual time with that memory. Two differences from the image and text instances of this pattern set the concrete form (Fig. 2). First, the conditioning is not a fixed-size embedding. It is an observation set of variable size K . Thus `Enc` is a transformer over per-observation tokens. Its output is a *memory* of K encoded states, not one pooled vector. `Dec` reads the memory by cross-attention, the one standard mechanism that does not depend on K . Second, the generated object is a short vector of d parameters, not an image. Thus the decoder is small. The d components of m are d tokens. Self-attention among them expresses posterior correlations, such as the ridge of the Fisher–KPP posterior or the drift–volatility coupling of GBM. The virtual time τ enters every block through zero-initialized adaptive layer normalization (adaLN-Zero), the conditioning mechanism of diffusion transformers (Peebles and Xie, 2023). The untrained network is then an identity-like map.

On the encoder side, a few features describe observation i , and a learned linear map lifts them to the model width. There are four features. The first is the time t_i . The second is the transformed value $w(v_i)$. Here v_i is the observed value in the coordinate the problem declares, and w is the learnable monotone map (Sec. 3.3). The third is an index of the sensor or state component that produced the value. A multivariate state gives one token per observed component per time. The fourth is the design-size feature of Sec. 3.3. The next subsection describes the attention mechanism that lets the tokens interact. Section 3.3 describes the choice of features.

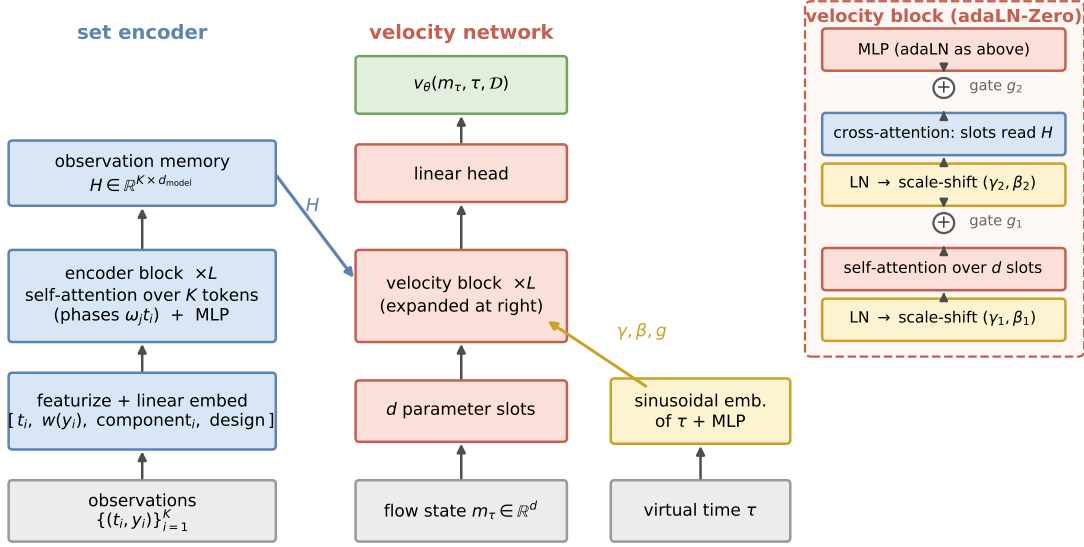


Figure 2: Model architecture. Arrows show the data flow, bottom to top. Left: each observation is featurized and linearly embedded into one token. L transformer encoder blocks produce a memory H of K encoded states. Their attention phases come from the physical observation times (Sec. 3.2). Middle: the d components of the flow state m_τ form d parameter-slot tokens. L velocity blocks and a linear head map them to the velocity v_θ . Right: one velocity block. It has self-attention across the parameter slots, cross-attention that reads the observation memory H , and an MLP. Layer normalization with scale-shift (γ, β) precedes each part, and a gated residual connection (g , marked $+$) follows it. A zero-initialized map produces the triples (γ, β, g) from the virtual time τ , so the untrained block is the identity (Peebles and Xie, 2023).

This scheme is the survivor of a sequence of measured alternatives. The failed alternatives stay in the package as reproducible controls. All sizes are configuration parameters. One configuration (width 64, four heads, three blocks per side) serves every experiment in this report. At the training-set sizes we use, models of this scale are sufficient: doubling width and depth changes no benchmark result. Nothing in the construction caps the scale.

3.2 Attention phases from observation times

We adapt rotary position embeddings, the standard positional scheme of modern transformers (Su et al., 2024), to irregular time sampling.

First the classical construction. Token states $x_1, \dots, x_K \in \mathbb{R}^{d_{\text{model}}}$ enter an attention layer. Each head computes

$$q_i = W_Q x_i, \quad k_i = W_K x_i, \quad u_i = W_V x_i, \quad W_Q, W_K, W_V \in \mathbb{R}^{d_h \times d_{\text{model}}}, \quad (5)$$

and token i aggregates the other tokens with weights $a_{il} = \text{softmax}_l(\hat{q}_i^\top \hat{k}_l / \sqrt{d_h})$. The hats mark position-modulated vectors. Each query and key is split into $d_h/2$ planar blocks $q_i^{(j)} \in \mathbb{R}^2$. Block j is rotated by an angle that a per-token scalar ϕ_i sets,

$$\hat{q}_i^{(j)} = R(\omega_j \phi_i) q_i^{(j)}, \quad \hat{k}_i^{(j)} = R(\omega_j \phi_i) k_i^{(j)}, \quad R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}. \quad (6)$$

In the classical scheme the scalar is the integer position of the token in the sequence, $\phi_i = n_i$. The frequencies form a geometric progression across blocks, $\omega_j = \omega_{\text{max}}(\omega_{\text{min}}/\omega_{\text{max}})^{(j-1)/(d_h/2-1)}$. High-frequency blocks resolve neighboring positions and low-frequency blocks resolve long ranges. Rotations preserve inner products up to the relative angle,

$$\hat{q}_i^\top \hat{k}_l = \sum_{j=1}^{d_h/2} (q_i^{(j)})^\top R(\omega_j(\phi_l - \phi_i)) k_l^{(j)}, \quad (7)$$

so every attention logit depends on the positions only through the differences $\phi_l - \phi_i$. This gives relative position awareness with no positional parameters.

For irregularly sampled data the integer position has no meaning. The generalization is direct. We set $\phi_i = t_i$, the physical time stamp of observation i . We choose the frequency range so that the periods $2\pi/\omega_j$ span from the finest observable time gap to several times the horizon. Nothing else changes. By (7), every logit now depends on the time stamps only through the differences $t_l - t_i$. Time-aware attention models for irregular clinical and event series make the same substitution in other positional schemes (Horn et al., 2020; Shukla and Marlin, 2021). Two consequences follow. The encoder is exactly permutation-invariant: relabeling the tokens leaves the right-hand side of (7) unchanged, and we verified this numerically to 10^{-8} . Variable designs need no positional bookkeeping. The phases enter only the self-attention of the encoder. The cross-attention from the parameter slots to the observation memory is unmodified attention. The parameter slots carry no time stamp, and the memory states already encode the temporal structure.

Integer positions create an implicit train/test contract on token placement. We trained a model on spread-out subsampled positions and evaluated it on contiguously packed ones. The posterior acquired substantial biases while every component stayed individually correct. With physical-time phases the contract does not exist.

3.3 Input features

The remaining design choices concern the content of the tokens. Four of them matter. The first is the coordinate of the observed value. The other three are learnable modules. The first is a transform of the observed values. The second is an optional increment-based token construction for directly observed diffusions. The third is an explicit design-size feature. Each module occupies a few lines of code. Each module repairs a measured failure in the conditioning of the numbers that the set encoder receives. Each repair is a learnable module and not a problem-specific transformation.

Value coordinate. The problem declares the coordinate in which the observed value enters the token: the value itself, or its logarithm. The package has no default for this choice. The rule is: use the coordinate in which the observation noise is additive. For log-normal assay noise this is the logarithm. The learnable transform of the next paragraph does not replace this rule. With raw values under log-normal noise, the populations below about 0.05 are compressed into an interval below the resolution of the network. The noise model measures their logarithm to ± 0.1 . On such observation sets the posterior centre then misses by several posterior widths. We measured this on the Lotka–Volterra system of the `sbibm` benchmark. The token value changed from the raw population to its logarithm. The median classifier two-sample test against MCMC improved from 0.79 to 0.73. All measurements in this report use raw values.

Value transform. The state variables of the benchmark systems live on very different scales. Some of them (prices, concentrations, populations) are multiplicative: the information sits in relative changes, and the natural coordinate is logarithmic. Ignoring this is costly. On GBM, raw price tokens under global standardization force the network to divide squared increments by squared levels internally. The result is a persistent volatility bias that training does not remove. We do not prescribe coordinates per problem. Each value enters its token through a learned monotone map,

$$w(x) = \sum_{k=1}^S a_k \operatorname{sign}(x) \log(1 + |x|/s_k), \quad a_k \geq 0. \quad (8)$$

Each basis function is linear for $|x| \ll s_k$ and logarithmic for $|x| \gg s_k$. The family contains the identity, the signed logarithm and every compression in between. The weights are positive, so w is monotone by construction. Every candidate compression is present from initialization. Training only reweights fixed nonlinearities, and the downstream network can use the helpful coordinate from the first optimizer step. The scales s_k sit on a fixed geometric grid that covers the dynamic range of the data. There are $S = 8$ octaves. One setting serves all fourteen systems, with no per-problem adjustment.

We compared this structure with simpler learnable alternatives on geometric Brownian motion. There the correct coordinate, the log-price, is known, so a failure to find it is visible against the closed-form posterior. A generic two-layer MLP on the raw values and a random-Fourier-feature map both fail to find the coordinate at a matched budget. A free-form map must discover the shape and the scale of the compression at the same time. Its gradients arrive through the whole downstream network. In practice

it does not find them. The Yeo–Johnson power transforms (Yeo and Johnson, 2000) also fail. This is a classical one-parameter family, in essence $x \mapsto ((1 + |x|)^\lambda - 1)/\lambda$ with sign symmetry. It passes through the identity at $\lambda = 1$ and tends to the logarithm as $\lambda \rightarrow 0$. Trained end-to-end, λ never leaves its initialization. A single global exponent must cross a gradient plateau before the downstream network learns anything that rewards the move. The parametrization decides the outcome, not the hypothesis class. The mixture (8) finds the coordinate reliably. The trained weights concentrate on the small-scale components: the model learns the log-price.

Increment features for directly observed diffusions. For scalar SDEs whose state is observed directly, with no latent components, the package offers a second token construction. After a sort by time, consecutive observations contribute

$$[t_{(i)}, w(y_{(i)}), \Delta w_i, (\Delta w_i)^2, w_t(\Delta t_{(i)})], \quad \Delta w_i = w(y_{(i+1)}) - w(y_{(i)}).$$

The motivation is elementary. For a Markov observed process the likelihood factorizes over consecutive pairs. Transition densities of diffusions depend on increments and gaps. Squared increments carry the quadratic-variation information that identifies diffusion coefficients. This is an inductive bias of the same kind as convolutions for images: useful where its assumption holds, not a requirement of the method. The point-token path of Sec. 3.1 is fully general.

The rule for the choice between the constructions is: use increment tokens exactly when the observed process is Markov. We tested the rule in both directions. An increment-token model transfers across Markov systems without adjustment. On the Heston model the observed price is not Markov, because the volatility is latent. There increment and point tokens perform identically, as the factorization argument predicts. A per-problem flag selects the construction.

Two normalization details matter for processes with jumps, where squared increments within one observation set span several orders of magnitude. A running feature normalization then takes its scale from the rare outliers and never converges. The representation drifts during training, the flow trains against a moving target, and the Merton jump-diffusion acquires a substantial volatility bias. Probes of the encoder show a failure of representation stability, not of extraction. The failing configuration encodes robust volatility statistics as well as the healthy one does. Two repairs follow. First, the increment features are logarithmically compressed before normalization. This is a near-identity on systems without jumps, since $\log(1 + x) \approx x$ for small x , and in our experiments it removes the bias. Second, a smaller residual remains. Its mechanism is a jump-classification threshold that floats with the unknown volatility. To remove it, the normalization adapts to the observation set. Robust summary statistics s of the set are pooled over the non-padding positions. A zero-initialized MLP maps them to per-channel scale and shift corrections,

$$h_c \mapsto (1 + \gamma_c(s)) h_c + \beta_c(s). \quad (9)$$

This is a feature-wise affine modulation, FiLM in the terminology of Perez et al. (2018). It removes the residual as well as a hand-crafted per-observation-set rescaling does. Three conditions are necessary for it to help: zero initialization, modulation applied *after* the running normalization, and pooled statistics that exclude the padding positions.

Design size. The number of observations governs the posterior width: in regular models the error scales as $K^{-1/2}$. Softmax attention computes normalized weighted means and is therefore almost invariant to the size of the set it aggregates. Probes of the encoder confirm the mechanism. Every variant we tested recovers posterior-mean statistics from the pooled output. How well the pooled output encodes $\log K$ varies widely across variants, and it ranks them exactly in the order of their width errors. We supply this one number as an explicit feature, $\log K$ in every token. This removes most of the width error at dense designs. The design-resampling training procedure of Sec. 4 removes the rest. A few learned register tokens (Darcet et al., 2024) give the same effect without an explicit feature.

4 Training procedures

Common settings. All measurements in this report sample with the midpoint scheme, 20 steps, on a uniform time grid. We fixed the step count with a convergence check on the benchmark suite: fourth-order Runge–Kutta with three times as many steps changes no benchmark statistic. That check holds for the posteriors of the suite, which are 4–107 times narrower than the prior at the median instances. It does not hold for posteriors that are much narrower. We found this later, on the Lotka–Volterra system

of `sbibm` with 20 observations, where the posterior is 20–280 times narrower than the prior. There, near $\tau = 1$, the Jacobian of the field is about $-1/(1 - \tau)$ until $1 - \tau$ is of the order of the posterior width, in the normalized coordinates. A uniform grid of n steps cannot resolve a posterior narrower than about $1/n$ with any explicit scheme. The samples then have the correct marginal widths and the wrong joint structure. Take the exact flow-matching field of a Gaussian target of width 0.01. The uniform grid with 20 midpoint steps gives a classifier two-sample test of 0.92 against the target (0.5 means not distinguishable). Fourth-order Runge–Kutta with 50 uniform steps gives 0.60. A grid whose steps shrink towards $\tau = 1$ as $(1 - i/n)^3$ gives 0.50 with the same 20 midpoint steps, down to a width of 0.003. On the trained Lotka–Volterra flow the same change moves the test from 0.80 to 0.57 at no extra cost. The package now has both grids, and the caller must name the grid, the scheme and the step count in every sampling call. Use the shrinking grid for posteriors that can be much narrower than the prior. Evaluation uses an exponential moving average of the training weights (decay 0.999). The optimizer is Muon (Jordan et al., 2024). A Newton–Schulz iteration orthogonalizes the momentum of every two-dimensional weight before the update. Adam updates the biases, normalizations and embeddings. Its step size follows a measured batch-size rule $\eta(B)$ (Appendix B). The schedule is cosine with linear warmup, and the gradients are clipped. At every optimizer step the virtual times are drawn afresh, stratified over $[0, 1]$: $\tau_i = (i - 1 + u_i)/B$ for a batch of size B , with $u_i \sim U[0, 1]$ i.i.d., $i = 1, \dots, B$. Fresh reference noise m_0 is drawn with them. No pairing of observation sets and virtual times repeats. For fixed-design problems the training set has n simulated observation sets, and training runs for a fixed number of epochs over it. The calibration-screen gallery of Appendix A uses $n = 40,000$ simulations and 35 epochs.

Design amortization. Training data are simulated once on a fine grid. Observation designs are then resampled *from the stored fine trajectories at every optimizer step*. The procedure has three parts. A controlled comparison on GBM, Ornstein–Uhlenbeck and FitzHugh–Nagumo fixed each part:

1. *Fresh designs at every step.* Each step re-draws the observation subset of each trajectory. Design memorization becomes impossible, and one simulation serves arbitrarily many designs. The alternative is one fixed design per trajectory. It closes the dense-design error at long budgets but overfits the design set. In our experiments it gives overconfident posteriors at sparse K .
2. *The design-size law.* K is drawn 50% log-uniform on $[K_{\min}, K_{\max}]$ and 50% uniform on the dense half of the range. Log-uniform sampling alone puts too little mass at large K and leaves dense designs under-trained. The mixture removes the effect.
3. *Budget scaling.* For higher accuracy, scale simulations and optimizer steps together at a constant number of epochs. In our experiments the residuals then decrease steadily with this single budget variable. Passes over reused observation vectors eventually destroy accuracy. This is why the scaling must be joint.

With this procedure, the posterior width matches the exact per-design posterior closely and uniformly over the whole range of design sizes.

Budgets and stopping. Step budgets have deliberate slack, and validation chooses the shipped weights. Training monitors a validation set, built by the same tools as the test evaluation sets from disjoint seeds. The final weights are replaced by the best monitored checkpoint only when two conditions hold. The improvement must exceed the resolution of the monitor. The monitored value must sit at least three floors above that resolution. A monitor that ranks values of its own resolution returns noise in both directions. With that margin, overshooting a budget costs time and never quality. The step budget is therefore not a per-system quantity to tune. Appendix B gives the measurements behind the margin, including the two failure modes of the naive best-checkpoint rule.

5 Evaluation methodology

We measure accuracy against reference posteriors computed on the same observed points. A number in this report always says how far the posterior of the network is from the posterior that the data imply. It never says how well the network satisfies a self-consistency condition.

```

# ---- training a design-amortized posterior -----
m_all, X_all = simulate(prior, n)      # full fine-grid trajectories, once
theta_best, F_best = None, inf
for step in range(T):                  # step budget T, set with slack
    m1, X = batch(m_all, X_all, B)     # stored trajectories, reused
    tokens = tokenize(X, fresh_designs(B)) # per trajectory: K from the mix
                                           # law, K random (time, sensor) points

    tau = stratified(B)                 # fresh virtual times, stratified on [0,1]
    m0 = randn(B, d)                   # fresh reference noise
    z = (1 - tau) * m0 + tau * m1
    loss = norm(v(z, tau, Enc(tokens))) - (m1 - m0) ** 2
    muon_step(loss, lr=eta(B))          # batch step rule, cosine schedule
    ema.update()
    if step % Delta == 0:               # score the EMA weights on validation
        F = monitor(ema, val)
        if F < F_best: theta_best, F_best = copy(ema), F
# ship the best checkpoint only when it clears the monitor's own resolution
if F - F_best > phi and F_best >= kappa * phi: return theta_best
return ema

# ---- inference: n posterior draws for one observation set S -----
C = Enc(tokenize(S))                   # encode once; any size, any placement
z = randn(n, d)                        # or the learned base, where enabled
for tau, h in grid(L, kind):           # L steps; kind = "uniform" or "late"
    z_mid = z + h/2 * v(z, tau, C)     # midpoint scheme, 2 evaluations per step
    z = z + h * v(z_mid, tau + h/2, C)
return denormalize(z)

```

Figure 3: Training (top) and inference (bottom). The step rule $\eta(B)$ is that of Sec. 4, the monitor period is $\Delta = T/24$, the resolution multiple is $\kappa = 3$ with monitor floor φ (Appendix B). The measurements of this report use $L = 20$ steps on the uniform grid. The caller names L , the scheme and the grid in every sampling call (Sec. 4).

5.1 Reference posteriors as the record

Where a tractable likelihood exists, we compare distributions on the same observed points. The headline statistic is one number per observation set. It is the squared Fréchet distance between Gaussian fits of the two sample sets. The coordinates are reference-normalized: each parameter is divided by the standard deviation of the reference posterior,

$$d_F^2 = \|\mu_a - \mu_r\|^2 + \text{tr}(\Sigma_a + \Sigma_r - 2(\Sigma_r^{1/2}\Sigma_a\Sigma_r^{1/2})^{1/2}). \quad (10)$$

This is the construction of the FID score in the image-generation literature, and we use that name. The quantity is dimensionless only after a choice of scale. We use two scales. Let $\sigma_{r,i}$ be the standard deviation of the reference posterior in coordinate i and Δ_i the width of the prior box. Then

$$\text{FID}_\sigma = d_F^2(\{m_a/\sigma_r\}, \{m_r/\sigma_r\}), \quad \text{FID}_\Delta = d_F^2(\{m_a/\Delta\}, \{m_r/\Delta\}). \quad (11)$$

In FID_σ every coordinate is divided by the standard deviation of the reference posterior. This is the default, and an unadorned FID in this report means FID_σ . In FID_Δ every coordinate is divided by the prior range. The two answer different questions: error relative to what the data determine, and error relative to the scale of the problem. A model of fixed absolute accuracy reads very differently in the two scales where the posterior is narrow (Sec. 7.3).

FID is zero for a perfect match. It is dimensionless. It combines the three ways a posterior can be wrong: location, width and orientation. For scale: a bias of 0.1 reference-sd in each of d dimensions gives $\text{FID} \approx 0.01d$. A posterior 30% too wide in one dimension gives $\text{FID} \approx 0.09$. Where the direction of a failure matters, we also quote the per-parameter decomposition. A posterior that is too wide is conservative. A posterior that is too narrow misleads without a warning. The decomposition gives the signed posterior-mean error in reference-sd units (*bias*). It also gives the ratio of standard deviations (*width*). A width of 1 is exact, above one is conservative, below one is overconfident.

The engines of the next paragraph build the references. Appendix C gives the construction. A second chain from a well-separated start validates every MCMC reference. A reference is used only when its

```

import torch
from amortix import FlowPosterior, DESIGN_ZOO

prob = DESIGN_ZOO["heston"]()          # or any DesignProblem subclass; the class
                                       # declares value_coord ("raw" or "log")
post = FlowPosterior(prob)             # embedding/roPE resolve from the class
post.fit(n_train=120_000, steps=72_000, # about 1 h on one GPU
         retokenize=prob.make_retokenizer()) # fresh designs at every step

gen = torch.Generator().manual_seed(0)
m, raw = prob.simulate(1, gen)         # here: one synthetic observation set
tidx, cidx = prob.sample_design(gen, 20) # 20 (time, sensor) points
tokens = prob.tokens_for(raw[0], tidx, cidx, gen)
draws = post.sample(tokens, n=2000, n_steps=20, solver="midpoint",
                   t_grid="late")      # [2000, d] draws, milliseconds

```

Figure 4: The interface for the procedure of Fig. 3: a benchmark system from the design zoo, training, and posterior draws for one observation set. The sampling settings have no default values. Appendix D maps the terms of this report to the package options.

two chains agree. The between-chain discrepancy of the posterior mean, in posterior-sd units, is the floor below which nothing can be measured. We quote that floor next to the value of the model throughout. The floor scales with the chain length. The Merton jump-diffusion needs 2×10^5 -draw chains to clear it: two chains then agree to 0.016 sd in the median and 0.060 at worst. A system whose chains do not converge at any affordable length is reported under the calibration screen instead. After the engine upgrades of the next paragraph, no system of the suite is in that state.

Reference samplers and their admission. For nonlinear ODEs and SDEs, trustworthy reference posteriors are a task in themselves. For every system we tested several approaches and adopted the per-problem methods below. The structure of the likelihood decides the sampler. Where a closed form exists, we draw the posterior exactly (linear-Gaussian; GBM through its conjugate structure). Where the likelihood is exact but the posterior is not, adaptive Metropolis suffices for the well-conditioned systems. These are Ornstein–Uhlenbeck, Cox–Ingersoll–Ross (through its noncentral χ^2 transitions), SEIR, Merton, pharmacokinetics and Fisher–KPP. FitzHugh–Nagumo and Hodgkin–Huxley have needle-thin or multimodal posteriors. Their sharpest evaluation instances concentrate the posterior to 0.4% and 1.0% of the prior range in the narrowest coordinate. Adaptive Metropolis chains from separated starts disagreed on them by hundreds of posterior standard deviations. These two systems use nested sampling with *dynesty* (Skilling, 2006; Speagle, 2020). On the median instances their posteriors are unremarkable (3–30 $\times$ narrower than the prior). The needle is a property of the sharp tail, and there chain methods fail. Double well and polynomial drift have no transition density in closed form. Their likelihood is an unbiased bridge estimate (Durham and Gallant, 2002) inside pseudo-marginal MH (Andrieu et al., 2010). Stochastic Lotka–Volterra and Heston are the two systems whose likelihood is available only as a Monte-Carlo estimate, not as a computable number. Nested sampling assumes a deterministic likelihood and does not apply to them. Their references come from adaptive tempered SMC (Del Moral et al., 2006) over an unbiased particle estimate, admitted by population doubling. The estimate enters a pseudo-marginal kernel, so its variance is a property of the reference. The log of an unbiased estimator is biased downward by about half its variance. Where that variance varies over the parameter space, the bias tilts the posterior. Both systems therefore run at a particle count at which the spread of the estimator is a fraction of a nat. We verify this by repeated estimates at fixed parameters.

Every reference passes three checks before use. Two independent runs must agree below 0.25 posterior sd. The likelihood must reproduce the simulator: on noiseless data its value at the generating parameters is zero to machine precision. The generating parameters must lie inside the posterior. The three checks are not redundant. Each one caught a wrong likelihood in this work that the other two passed. The third check catches a reference that describes a different model than the simulator. Across the frozen evaluation sets of the suite, the generating parameters sit at a median 0.5–0.8 posterior sd from the reference mean. Only 0–1% of the coordinates lie beyond three (Appendix B).

Heston is the sharpest case of the third check, because the first two checks do not see the failure. We first built its reference from the idealized transition of the continuous-time model. That is a different

system	type	d	obs.	design	reference posterior
linear-Gaussian	linear map	4	full	variable	exact draws, no MCMC
geometric Brownian motion	SDE	2	full	variable	conjugate closed form
Ornstein–Uhlenbeck	SDE	2	full	variable	exact-likelihood MCMC
Cox–Ingersoll–Ross	SDE	3	full	variable	exact-likelihood MCMC
double well	SDE	3	full	variable	bridge, pseudo-marginal MH
polynomial-drift SDE	SDE	5	full	variable	bridge, pseudo-marginal MH
stochastic Lotka–Volterra	2-d SDE	4	both, pointwise price only	variable	bridge, tempered SMC
Heston	2-d SDE	5		variable	particle filter, tempered SMC
Merton jump-diffusion	jump SDE	5	full	variable	Poisson-mixture MCMC
FitzHugh–Nagumo	2-d ODE	4	v only	variable	exact-likelihood nested sampling
Hodgkin–Huxley	4-d ODE	4	V only	variable	exact-likelihood nested sampling
SEIR	4-d ODE	5	I, D	variable	exact-likelihood MCMC
pharmacokinetics (Bateman)	ODE (closed form)	3	concentration	variable	log-normal MCMC
Fisher–KPP	PDE (64-point grid)	2	3 sensors	variable	PDE-likelihood MCMC

Table 1: The benchmark suite. “obs.” names the observed coordinate where the state is only partly visible. Every system is trained and evaluated in the design-amortized mode, so each observation set carries its own (time, sensor) points. The last column names the instrument that builds the reference posterior of that system. The structure of the likelihood decides the choice (Sec. 5.1).

process from the one that generated the data. The simulator integrates the variance by clamped Euler. Past the Feller condition the variance spends real time pinned at zero. The discrepancy grows with the vol-of-vol ξ . On instances generated with $\xi \approx 0.9$ the reference placed ξ near 0.27 at half the true width. The two independent chains agreed on that wrong answer to 0.17 sd. Only the generating parameter, 8.1 posterior sd outside its own reference, exposed the defect. We replaced the idealized transition by a bootstrap particle filter that runs the update rule of the simulator itself. Two closed forms in the $\xi = 0$ limit admit the filter. The worst deviation moved to 2.6 sd, the two-chain agreement tightened to 0.06 sd, and the floor of the set halved. Against the corrected reference, the dependence of Heston on model size is unremarkable (Sec. 7).

Resolution of the estimator. Both sample sets are finite. Thus (10) is positive even for a perfect posterior, and the size of that offset sets what the metric can resolve. We measured it directly. We drew *two independent sample sets from the same exact GBM posterior* and scored one against the other. The median FID is 0.0157 at 500 samples, 0.0038 at 2,000, 0.0009 at 8,000 and 0.0002 at 32,000. This is a clean $1/n$, as the estimation error of the fitted means and covariances predicts. Two consequences follow. Differences below ≈ 0.005 at $n = 2,000$ (≈ 0.002 at the 4,000 draws that score the evaluation sets of Table 2) are not resolved. We report values in that range as “at the resolution limit” and do not compare them. The best values in this report sit above the limit, not at it. The design-amortized GBM model reads 0.0107 at $n = 2,000$, 0.0100 at 8,000 and 0.0072 at 32,000 on the same evaluation set. After subtraction of the estimator offset, its excess over the reference is ≈ 0.007 at every sample size. This is a real error of the model, not an artifact of the finite sample count.

Equation (10) is the squared 2-Wasserstein distance between Gaussian fits. The division by the reference standard deviation is ours, because the parameters carry different physical units. Distribution-free discrepancies and the classifier two-sample test of Lueckmann et al. (2021) rank the systems identically. The choice with consequences is the normalization, and the notation above makes it explicit.

6 Benchmark suite

The suite has fourteen systems (Table 1). This section defines each system with its equations, parameter ranges and observation model. Every system is trained and evaluated in the variable-design mode, with the observation subsets resampled at every optimizer step. This mode enables inference at any design, and the resampling is also a data augmentation over designs.

Euler–Maruyama integrates all SDEs on the stated fine grid. RK4 integrates the ODEs (Hodgkin–Huxley uses a semi-implicit scheme for the gating variables). Finite differences integrate the PDE. Every system is trained and evaluated in the design-amortized mode of Sec. 2. The observation times of every observation set are random, and so are the sensor indices for the PDE. One trained network serves all of them. The *classical estimator* listed for each system is the standard non-Bayesian method a practitioner would apply. All of them return point estimates, not distributions. They serve as an accuracy floor for the posterior mean. Three of them recur, and we define them here once. *Kramers–Moyal regression*

estimates the drift and the squared diffusion from the conditional first and second moments of the observed increments ($f(x) \approx \mathbb{E}[\Delta X \mid X = x]/\Delta t$, $\sigma^2(x) \approx \mathbb{E}[\Delta X^2 \mid X = x]/\Delta t$). It fits the parametric form to these estimates by least squares. *Multi-start nonlinear least squares* fits the deterministic solve of the system to the observed trajectory. It minimizes the squared error over the parameters. Restarts from many random initial points escape the local minima. *Exact maximum likelihood* is available where the transition density is known in closed form. Fixed-design observations are token sets over the union of the stated channels. “Fast/slow” channels observe every step over a short window and every j -th step over the horizon. Priors are uniform on the stated boxes. The token of every system carries the observed value itself (Sec. 3.3).

Every system runs in the design-amortized mode. An observation set draws K random (time, channel) points on the stated simulation grid. K lies between the per-system minimum and 128, under the design-size law of Sec. 4. The “channels” entries below also define the conventional fixed grids of the calibration-screen gallery of Appendix A, for the systems that appear there.

Linear-Gaussian. $y = Am + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.5^2 I_6)$, with a fixed correlated design matrix $A \in \mathbb{R}^{6 \times 4}$; $m \in [-3, 3]^4$. The posterior is exactly Gaussian. The classical estimator is the posterior mean itself. This system checks the flow engine in isolation from the encoder.

Geometric Brownian motion. $dS = \mu S dt + \sigma S dW$, $S_0 = 1$; $dt = 0.01$, 500 steps (five “years”); channels: fast (1×60), slow (10×40). Prior $\mu \in [-0.2, 0.4]$, $\sigma \in [0.1, 0.6]$. Classical: exact MLE on log-returns. Reference: conjugate posterior (Appendix C).

Ornstein–Uhlenbeck. $dX = -\theta X dt + \sigma dW$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$; $dt = 0.02$, 500 steps; channels: fast (1×50), slow (20×24). Prior $\theta \in [0.3, 3]$, $\sigma \in [0.2, 1.5]$. Classical: exact MLE from per-gap Gaussian transitions.

Cox–Ingersoll–Ross. $dX = a(b - X) dt + \sigma \sqrt{X} dW$; $dt = 0.02$, 500 steps; channels as OU. Prior $a \in [0.3, 3]$, $b \in [0.3, 1.5]$, $\sigma \in [0.1, 0.5]$. Classical: Euler pseudo-likelihood MLE (Cox et al., 1985).

Double well. $dX = (\theta_1 X - \theta_2 X^3) dt + \sigma dW$; $dt = 0.01$, 1000 steps, with noise strong enough for barrier crossings; channels: fast (1×80), slow (25×36). Prior $\theta_{1,2} \in [0.5, 3]$, $\sigma \in [0.4, 1.2]$. Classical: Kramers–Moyal regression (defined above).

Polynomial-drift SDE. $dX = (c_0 + c_1 X + c_2 X^2 + c_3 X^3) dt + \sigma dW$. This is drift identification on a dense polynomial basis, the setting of SINDy (Brunton et al., 2016) without its sparsity prior. Sec. 8 discusses a sparse-coefficient variant; $dt = 0.01$, 1000 steps; channels: fast (1×80), slow (20×45). Prior $c_0, c_2 \in [-0.5, 0.5]$, $c_1 \in [-2, 0]$, $c_3 \in [-1, -0.1]$, $\sigma \in [0.3, 0.8]$. Classical: least squares of the Kramers–Moyal drift estimate on the polynomial basis, the dense-basis version of SINDy regression.

Stochastic Lotka–Volterra. A prey/predator pair with multiplicative noise, $dx_1 = (\alpha x_1 - \beta x_1 x_2) dt + 0.05 x_1 dW_1$, $dx_2 = (\delta x_1 x_2 - \gamma x_2) dt + 0.05 x_2 dW_2$; $dt = 0.01$, 600 steps. Each observation token is one (time, species) point, so a design of K points spreads over both species; $K \in [4, 128]$. Prior $\alpha, \gamma \in [0.8, 1.5]$, $\beta, \delta \in [0.4, 1.2]$. Classical: multi-start nonlinear least squares (defined above).

Heston. $dS = \mu S dt + \sqrt{v} S dW_1$, $dv = \kappa(\bar{v} - v) dt + \xi \sqrt{v} dW_2$, $\text{corr}(dW_1, dW_2) = \rho$ (Heston, 1993). The price is observed and the volatility is latent; $dt = 0.01$, 500 steps, designs of $K \in [2, 128]$ random times. Prior $\mu \in [-0.1, 0.3]$, $\kappa \in [0.5, 5]$, $\bar{v} \in [0.02, 0.3]$, $\xi \in [0.1, 1]$, $\rho \in [-0.9, 0]$.

Merton jump-diffusion. $dS/S = \mu dt + \sigma dW + d(\sum_i (e^{J_i} - 1))$ with Poisson jump rate λ and $J_i \sim \mathcal{N}(\mu_J, s_J^2)$ (Merton, 1976); $dt = 0.01$, 500 steps, $K \in [2, 128]$. Prior $\mu \in [-0.1, 0.3]$, $\sigma \in [0.1, 0.5]$, $\lambda \in [0.2, 3]$, $\mu_J \in [-0.3, 0.3]$, $s_J \in [0.05, 0.4]$. Reference: Poisson-mixture transition density (Appendix C).

FitzHugh–Nagumo. The classical two-dimensional reduction of the neuronal excitability equations (FitzHugh, 1961; Nagumo et al., 1962): $\dot{v} = v - v^3/3 - w + I$, $\dot{w} = \epsilon(v + a - bw)$. Only v is observed, at arbitrary times drawn per observation set ($K \in [4, 64]$, horizon 40, $dt = 0.05$), with observation noise 0.05. The recovery variable w is never seen. Prior $a, b \in [0.5, 0.9]$, $\epsilon \in [0.05, 0.3]$, $I \in [0, 0.5]$. This is deliberately harder than the toy configuration of the community, for example the three-parameter, fully

observed variant that ships with **pints** (Clerx et al., 2019). Partial observation leaves a latent coordinate. The drive I is a fourth unknown. The posteriors have the needle-and-mode geometry of Sec. 5.1. Classical: multi-start nonlinear least squares (defined above).

Hodgkin–Huxley. The standard four-dimensional membrane model (Hodgkin and Huxley, 1952). The voltage is observed at random times. The parameters are (g_{Na}, g_K, g_L, I) with prior $[60, 180] \times [10, 50] \times [0.05, 0.5] \times [4, 12]$; 3000 integration steps, $K \in [4, 96]$.

SEIR with mortality. A five-compartment stochastic epidemic (two-phase contact rate $\beta_1 \rightarrow \beta_2$, incubation α , recovery γ_r , mortality γ_d). The infected and deceased compartments are observed. Each token is one (time, compartment) point, $K \in [4, 64]$; 60-day horizon ($dt = 0.1$), observation noise 0.004. Prior $\beta_1 \in [0.2, 0.6]$, $\beta_2 \in [0.05, 0.3]$, $\alpha \in [0.2, 0.5]$, $\gamma_r \in [0.05, 0.2]$, $\gamma_d \in [0.005, 0.05]$. Classical: nonlinear least squares on (I, D) .

Pharmacokinetics. The standard one-compartment oral-dosing model of pharmacokinetics (e.g. Gabrielson and Weiner, 2000): absorption at rate k_a , elimination at rate k_e , distribution volume V . The Bateman concentration curve is $c(t) = \frac{D_0 k_a}{V(k_a - k_e)} (e^{-k_e t} - e^{-k_a t})$ with dose $D_0 = 500$; log-normal assay noise (10%); horizon 24 h, $K \in [3, 64]$ irregular draws. Prior $k_a \in [0.5, 4]$, $k_e \in [0.05, 0.4]$, $V \in [20, 100]$. Reference: log-normal likelihood MCMC. The measurements of this report put the raw concentration into the token. The rule of Sec. 3.3 says to use its logarithm for this noise model, and the package now requires the problem to declare that choice.

Fisher–KPP. $\partial_t \theta = D \partial_{xx} \theta + r \theta(1 - \theta)$ on $[0, 1]$ with reflecting boundaries, a 64-point finite-difference grid and a Gaussian initial bump. Three fixed sensors at $x = 16/63, 32/63, 48/63$ are observed at random (time, sensor) pairs with noise 0.02; $dt = 0.01$, 200 steps, $K \in [4, 96]$. Prior $D \in [0.001, 0.01]$, $r \in [2, 10]$. Reference: Gaussian observation model around the deterministic solve. One likelihood evaluation is one PDE solve.

7 Experimental results

This section is an experimental study. First we fix the protocol: references, baselines, model sizes and budgets. Then we give the main results: posterior overlays and the global metric on every system. Then we evaluate design amortization and the cost of training and inference.

7.1 Experimental protocol

References and baselines. Every accuracy number in this section is the FID of Sec. 5.1 against a reference posterior computed on the same observed points. We report the median over the evaluation set of the system. The evaluation set is a frozen collection of observation-set instances. It has 200 instances where the reference is closed form, 100 where it is cheap MCMC, and 32 where each reference costs minutes of validated MCMC. It is built once with two-chain validation (Sec. 5.1) and reused for every number in this report. Thus any two entries differ only in the model that is scored. The reference engines span more than six orders of magnitude in per-instance cost. The closed-form GBM posterior costs <1 ms. Exact-likelihood MCMC costs seconds. Nested sampling and pseudo-marginal chains cost minutes, and one hour on the hardest instances. Table 3 in Sec. 7.4 gives the full accounting, next to the training and inference cost of the amortized model. We do not report accuracy comparisons with other amortized packages. Their estimators consume fixed-length summary vectors. At package defaults we could not get converged posteriors from them on raw irregular time series. Fair problem-specific summary networks for every system are outside the scope of this report. The validated reference posteriors are the accuracy record throughout, not another package. Simulation-based calibration is the screen for the fixed-design gallery systems without a validated reference. Appendix A gives its per-parameter results.

Models and training. For each reference system we train the same architecture at three sizes: tiny (0.1M parameters), the default small (0.5M) and big (2.3M). Two miniature sizes, nano (26k) and pico (7k), probe the low-capacity end. All runs follow the procedure of Sec. 4. Simulation budgets range from 5,000 to 480,000 per system. The size sweep under the default recipe opens the results (Table 2). Observation designs are random at training and at evaluation wherever the system defines a design

distribution. Fixed designs appear only where a reference requires them. Every number of this section samples with 20 midpoint steps on the uniform grid and 4,000 draws per observation set (Sec. 4).

7.2 Main comparison: quality, model size, data, and baselines

One comparison, run identically on every task, opens the results. The recipe is the package recipe of Sec. 4: Muon with the batch step rule, batch 512, a deliberately generous 90,000 steps with validated stopping. We train it at five model sizes on every system with a validated random-design reference. The top three sizes have three seeds. We score the models on the frozen evaluation sets. Every system of the suite has a row. Every row carries a validated reference built by the engine that matches its likelihood (Sec. 5.1). These are closed forms where they exist. They are adaptive Metropolis or nested sampling where the likelihood is exact. They are pseudo-marginal MH or tempered SMC over unbiased estimates where the transition density has no closed form. Hodgkin–Huxley was built twice, once by tempered SMC and once by nested sampling. The two size sweeps agree to less than the seed spread at every model size. This is the strongest cross-check in this report. Table 2 is the result. Fig. 6 shows the corresponding posteriors.

system	floor	FID _σ at model size (parameters)				
		pico (7k)	nano (26k)	tiny (98k)	small (0.5M)	big (2.3M)
linear-Gaussian	0.0030	0.0660 ± 0.0131	0.0148 ± 0.0003	0.0068 ± 0.0009	0.0045 ± 0.0001	0.0042 ± 0.0005
geometric Brownian motion	0.0018	0.0064	0.0039	0.0036 ± 0.0004	0.0028 ± 0.0001	0.0035 ± 0.0006
Ornstein–Uhlenbeck	0.0013	0.0297	0.0281	0.0294 ± 0.0026	0.0282 ± 0.0031	0.0281 ± 0.0017
Cox–Ingersoll–Ross	0.0028	0.0786 ± 0.0178	0.0823 ± 0.0131	0.0733 ± 0.0110	0.0809 ± 0.0071	0.0700 ± 0.0036
pharmacokinetics	0.0023	0.0846	0.0319	0.0340 ± 0.0021	0.0262 ± 0.0062	0.0251 ± 0.0024
Merton jump-diffusion	0.0065	0.0663	0.0299	0.0219 ± 0.0007	0.0247 ± 0.0016	0.0267 ± 0.0050
Fisher–KPP	0.0100	0.0705 ± 0.0161	0.0415 ± 0.0033	0.0377 ± 0.0054	0.0307 ± 0.0036	0.0282 ± 0.0035
FitzHugh–Nagumo	0.0052	0.4495 ± 0.0596	0.2367 ± 0.1194	0.0864 ± 0.0394	0.0357 ± 0.0030	0.0406 ± 0.0265
SEIR	0.0069	0.1511 ± 0.0354	0.0642 ± 0.0112	0.0185 ± 0.0024	0.0114 ± 0.0027	0.0097 ± 0.0015
double well	0.0028	0.0201 ± 0.0047	0.0105 ± 0.0000	0.0089 ± 0.0009	0.0093 ± 0.0009	0.0139 ± 0.0020
polynomial-drift SDE	0.0055	0.0342 ± 0.0043	0.0200 ± 0.0045	0.0205 ± 0.0034	0.0271 ± 0.0041	0.0386 ± 0.0113
Heston	0.0052	0.1829 ± 0.0537	0.0753 ± 0.0107	0.0457 ± 0.0007	0.0508 ± 0.0044	0.0543 ± 0.0053
Hodgkin–Huxley	0.0049	9.76 ± 1.41	5.83 ± 0.87	2.37 ± 0.32	1.52 ± 0.46	0.313 ± 0.034
stochastic Lotka–Volterra	0.0073	7.29 ± 0.23	3.23 ± 0.58	1.42 ± 0.20	0.80 ± 0.10	0.95 ± 0.23

Table 2: The main result: every system of the suite against its validated reference. Entries are FID_σ, the median over the frozen evaluation set of Sec. 7.1. $K = 20$, Merton $K = 56$, Fisher–KPP $K = 40$. Fisher–KPP uses 4,000-draw PDE-likelihood chains, hence its higher floor. Linear-Gaussian references are exact, no MCMC. Each cell is one design-amortized network trained with the package recipe at 120,000 simulations. $\pm$ is the half-range over independent training seeds (single-seed cells at pico and nano carry none). The best cell of each row is in bold.

On GBM every size from tiny up sits within two floors of the resolution of the evaluation set (0.0028–0.0036 against 0.0018). Linear-Gaussian, double well and SEIR land within a few floors at the larger sizes. Those cells bound the error from above and do not measure it. Ornstein–Uhlenbeck (~ 0.028) and Cox–Ingersoll–Ross (0.07–0.08) do not depend on the model size. Their residuals belong to the task, not to capacity or recipe. FitzHugh–Nagumo and Hodgkin–Huxley improve most strongly with model size (0.45 $\rightarrow$ 0.036 from pico to small, 9.8 $\rightarrow$ 0.31 across the five sizes, with the largest model still improving). They carry the largest residuals of the suite. The stochastic Lotka–Volterra residual (0.80) sits almost entirely in one coordinate (Fig. 5). Capacity does not explain it (0.95 at the big size). Validated stopping has a measurable effect. Under the margin rule of Sec. 4, the best-on-validation checkpoint replaced the final weights in half of the 194 training runs behind Table 2. It did so in 13 of 15 runs on Cox–Ingersoll–Ross and in 15 of 15 on FitzHugh–Nagumo (Appendix B).

Figure 6 shows the object the package produces, the joint posterior itself. It shows the median-error test instance of every reference-block system, model against reference. The recipe and the checkpoints are those of Table 2.

7.3 Amortization over observation designs

The central claim of the package is that one trained network answers $p(m \mid S)$ for *every* observation design S , at any number and placement of points. Fig. 7 tests the claim on the six systems of the suite where a per-design reference can be computed and validated. Each test observation set draws a fresh random design, the reference posterior is recomputed on exactly those points, and FID is evaluated per

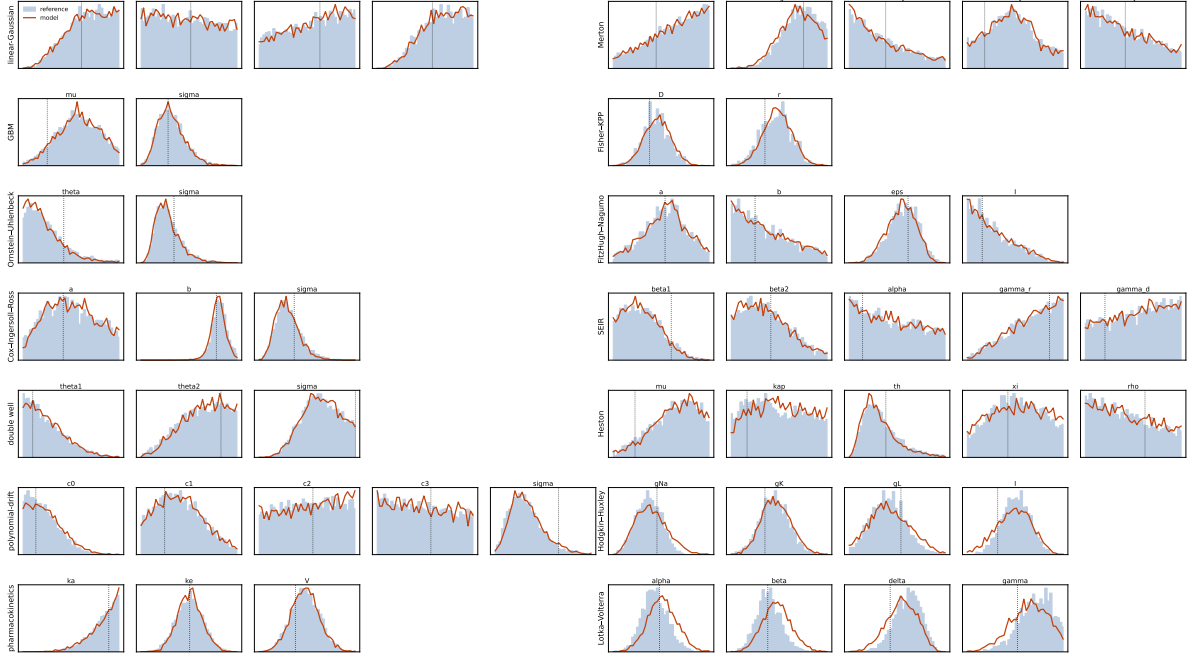


Figure 5: One-dimensional posterior marginals, model (line) against reference (filled), for every system of Table 2. Each panel shows the median-FID $_{\sigma}$ test instance of its system, with 4,000 draws per side. The dotted line is at the generating parameter. The axis of each panel spans its sample range, not the prior box (Fig. 6 shows the prior-relative scale). Rows are systems, columns their parameters. This is the same comparison that the table reduces to one number per cell. It shows what that number cannot. The model is exact on linear-Gaussian, GBM and double well. On other systems it is faithful but slightly over-dispersed. On Lotka–Volterra’s β and on Hodgkin–Huxley’s conductances on sharp instances, a single coordinate carries the whole error.

design-size bucket. In our experiments, amortization over designs is not a trade-off against accuracy. The stars in Fig. 7 are networks trained on a single fixed design. Wherever such a network exists, the amortized network is comparable to it at that design size. Only the amortized network answers anywhere else on the K axis. It also removes a failure mode, because the resampling of the observation subset at every optimizer step is also a data augmentation. The two systems with elevated FID are not noise. The next paragraph analyzes them.

Why FID $_{\sigma}$ rises with the design size. FID $_{\sigma}$ is a ratio. On pharmacokinetics ($0.07 \rightarrow 0.34$ from $K = 5$ to 50, Fig. 7) only its denominator moves. We measured this on the same evaluation sets, in prior-range units. The absolute posterior-mean error of the network is flat in K ($\varepsilon_{\Delta} = 1\text{--}12 \times 10^{-3}$, no trend). The reference posterior contracts by the $\sqrt{K}$ of a regular model (widths fall $2.8\text{--}5.1\times$, against $\sqrt{10} \approx 3.2$). The bias term $\sum_i (\varepsilon_{\Delta,i}/w_{\Delta,i})^2$ grows accordingly, $0.025 \rightarrow 0.121$. In the prior normalization the same model reads $\text{FID}_{\Delta} \approx 3 \times 10^{-4}$ at the densest designs. The posterior widths stay within 1–8% of the reference at every K . The network has a fixed localization floor of $10^{-3}\text{--}10^{-2}$ of the prior range. What grows is its visibility, once the data determine the parameters more precisely than that. We therefore report the pharmacokinetic entries per design size and do not pool them. The narrowest-posterior instances read several times worse than the rest. A median over one 32-instance evaluation set moves between 0.04 and 0.15 between otherwise identical training runs. The stochastic systems stay an order of magnitude above the floor even at $K = 100$. Fisher–KPP approaches the floor only at its densest bucket (0.126 at $K = 90$).

7.4 Cost of training and inference

Training under the package recipe (90,000 steps at batch 512, 120,000 simulations, validated stopping) costs 58–82 minutes per system at the small size on one H200 GPU. The big size costs 67–111 minutes. Hodgkin–Huxley costs 136 minutes, because its simulator is the most expensive. The budget is deliberately generous. With the checkpoint chosen by validation, overshooting it costs time and never quality (Sec. 4). At a 20,000-simulation budget training takes 4–8 minutes. The same run on a laptop CPU takes

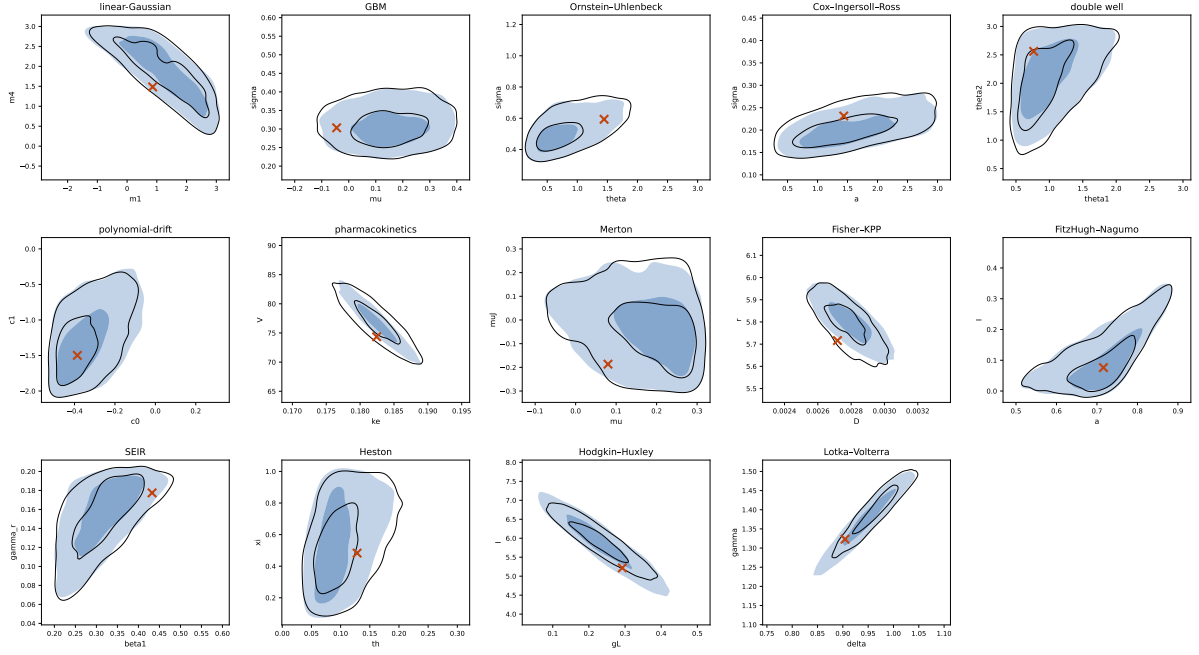


Figure 6: Joint posteriors of **amortix** (filled contours: 39% and 86% highest-density regions) against exact or validated reference posteriors (open contours, same levels). Each panel is the median- FID_σ test instance of one system of Table 2, from the big model of the package recipe, with 4,000 draws per side. The cross marks the true parameters. Each panel shows the most correlated parameter pair of that system. The comparison is thus of the shape of the dependence and not only of the marginal spread. Axes span the sample range of each panel, not the prior box. The prior is $4\text{--}107\times$ wider than the posterior standard deviation across the panels shown, and the sharp FitzHugh–Nagumo and Hodgkin–Huxley instances of Sec. 5.1 reach $\sim 230\times$. The plotted shapes carry no information about absolute sharpness.

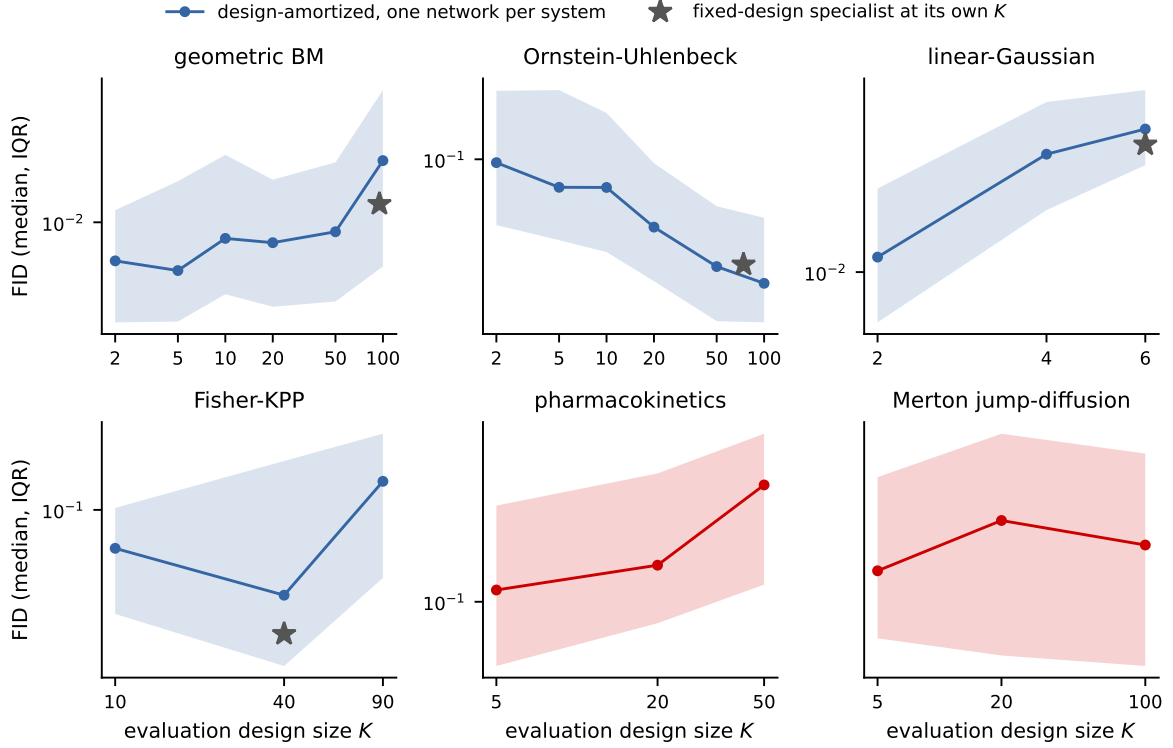


Figure 7: FID as a function of the evaluation design size, for one design-amortized network per system. The line is the median and the band is the interquartile range over the test observation sets of each bucket. A bucket has 100 sets for GBM, OU and linear-Gaussian, and 32 for the MCMC-referenced systems. Each set has a fresh random design and a per-design reference recomputed on exactly those points. These are the six systems whose references were built on a grid of design sizes. The rest of the suite has references at one design size (Table 2), which scores a model but does not trace a curve. Stars: networks of the same architecture trained on a single fixed design. They exist only at their own K and are shown for context. The Merton and pharmacokinetics curves use the converged references of Sec. 5.1. The text analyzes the pharmacokinetic rise with K , which is the precision floor. Both axes are logarithmic. Note the different vertical scales.

engine	systems	one posterior
closed form / conjugate	linear-Gaussian, GBM	<1 ms
Poisson-mixture / log-normal MCMC	Merton, pharmacokinetics	5–7 s
exact-likelihood MCMC	OU, CIR	6–34 s
tempered SMC, particle-filter likelihood	Heston	58 s
nested sampling (dynesty)	FitzHugh–Nagumo	≈2 min
PDE-likelihood MCMC	Fisher–KPP	186 s
tempered SMC, bridge likelihood	stochastic Lotka–Volterra	≈4 min
exact-likelihood MCMC	SEIR	≈16 min
nested sampling (dynesty)	Hodgkin–Huxley	≈18 min
pseudo-marginal MH, bridge likelihood	double well, polynomial-drift	≈38 min
amortix small (GPU), after training	any of the above	0.07 s
amortix small (laptop CPU)		0.43 s

Table 3: Per-instance cost of one reference posterior at the settings that built the frozen evaluation sets. MCMC rows are single chains of $1.5\text{--}2 \times 10^5$ draws (Fisher–KPP 4,000; double well and polynomial drift 60,000 pseudo-marginal draws), thinned to the 4,000 stored draws. The two-chain admission of Sec. 5.1 runs each engine two to three times. Every engine runs on the CPU, with one to a few threads on one server core. The exception is the Heston SMC filter. It runs on the GPU at the production population of 16,384 θ -particles over a 4,096-particle filter (median over 96 runs, spread 28–144 s). The CPU chains are single-instance timings. The dynesty rows are medians over the 32 test instances, reconstructed from the per-job wall times of the reference builds (`nlive=1000`). The spread across instances is about four on FitzHugh–Nagumo and about twenty on Hodgkin–Huxley. The hardest Hodgkin–Huxley instances take one hour. Training the amortized network costs 58–82 minutes on one H200 GPU (text above). After training, one posterior costs the entry in the last two rows.

about four times longer. It gives a bit-identical posterior, because all stochastic generation stays on a CPU random number generator. At inference, 2,000 posterior samples for one observation set cost 72 ms batched (135 ms single) on the GPU. On the laptop CPU they cost 426 ms. This holds at any design size and placement.

Table 3 sets these numbers against the per-instance cost of the reference engines that built the evaluation sets, at the settings that built them. The chains are long enough to clear the two-chain floors of Sec. 5.1. This is reference grade and not user grade: a 2,000-draw exact-likelihood chain costs 42–124 ms. The comparison cuts both ways. Where the likelihood costs microseconds, exact-likelihood MCMC gives a user-grade posterior in tens of milliseconds and a reference-grade one in seconds. That is cheaper than the training of the amortized network for any realistic number of observation sets. This is why those engines are the accuracy record of this report, not the network. The relation inverts where one likelihood evaluation is itself a simulation. A Fisher–KPP reference costs 186 s (a PDE solve per evaluation). A SEIR reference costs 16 minutes (an epidemic integration per evaluation). A Hodgkin–Huxley reference costs a median 18 minutes of nested sampling. A double-well or polynomial-drift reference costs 38 minutes of pseudo-marginal MH. At those prices the entire training run amortizes within a handful of posteriors. The number is two on the double well, four on SEIR and on Hodgkin–Huxley, and about twenty on Fisher–KPP. Every posterior after that costs 70 ms. Sequential refinement and design search require posteriors by the hundred. That is the intended regime of the package.

8 Discussion and limitations

The evidence supports one specific claim. For dynamical-systems inverse problems, the input representation of an amortized posterior estimator is the dominant accuracy factor, not a preprocessing detail. It can be part of the model and not part of the problem specification. With prescribed coordinates our own estimator acquires the failures of Sec. 3.3. The learnable representation removes them with one configuration for all fourteen systems. The reference posteriors of Sec. 5 certify the result.

Limitations. Bayesian experimental design, the intended consumer of design amortization, is not demonstrated yet. The training procedure of Sec. 4 makes the required object $p(m \mid S)$ coherent across subsets S . No real-data study with posterior predictive checks is included. Priors are limited to factorized distributions with tractable marginal CDFs, the class the probit map (2) covers. Correlated priors need a different normalization. Priors with atoms at zero, which would admit a sparse-coefficient variant of the polynomial-drift system, are a planned extension. The two hardest systems are not at reference quality.

system	per-parameter SBC p -values					
linear-Gaussian	m_1 : 0.06	m_2 : 0.60	m_3 : 0.38	m_4 : 0.56		
Ornstein-Uhlenbeck	θ : 0.28	σ : 0.77				
SEIR	β_1 : 0.33	β_2 : 0.22	α : 0.61	γ_r : 0.34	γ_d : 0.53	
geometric Brownian motion	μ : 0.26	σ : 0.44				
Cox-Ingersoll-Ross	a : 0.65	b : 0.001	σ : 0.73			
double well	θ_1 : 0.78	θ_2 : 0.54	σ : 0.88			
stochastic Lotka-Volterra	α : 0.07	β : 0.69	δ : 0.08	γ : 0.35		
FitzHugh-Nagumo	a : 0.19	b : 0.15	ϵ : 0.90	I : 0.63		
polynomial-drift SDE	c_0 : 0.06	c_1 : 0.96	c_2 : 0.46	c_3 : 0.80	σ : 0.50	

Table 4: Per-parameter SBC uniformity p -values for the standard-budget fixed-design gallery run (500 observation sets $\times$ 200 draws; $p > 0.05$ passes). The single failure is in bold.

Hodgkin-Huxley still improves at the largest model size. Capacity does not explain the Lotka-Volterra residual (Sec. 7.2). That residual is the clearest open problem in the suite. One more limitation concerns the sampler. The uniform time grid of these measurements does not resolve posteriors much narrower than the prior (Sec. 4). The numbers of this report are not affected, because the posteriors of the suite are wide enough. Numbers on sharper problems measured with that grid are lower bounds on the quality of the model.

A Controls and calibration screen

Controls. Two model-free controls accompany every result. The *prior-mean predictor* ignores the data (exactly 25% mean absolute error under a uniform prior). The *ridge control* is a point predictor of the parameters: a degree-2 ridge regression on generic summary statistics of the token set. Per token feature it uses the mean, the standard deviation, the mean square, the mean absolute value and the 0.1/0.5/0.9 quantiles. These are computed over the observations that are present. It recovers no distribution and is not meant to. Its role is to detect one silent failure that the calibration screen cannot see. A posterior that ignores the data and returns the prior passes SBC perfectly. Only a point-accuracy control unmasks it.

Calibration of the fixed-grid gallery. Table 4 lists the per-parameter SBC p -values of the standard-budget gallery run (40,000 simulations). The protocol: for each system, 500 synthetic observation sets are simulated with known parameters. For each observation set the model gives 200 posterior draws, and the rank of the true parameter among the draws is recorded. A calibrated posterior gives uniform ranks. Each p -value tests this uniformity for one parameter. All parameters pass except Cox-Ingersoll-Ross b . Its deviation corresponds to a location shift of the posterior in b of ≈ 0.15 sd, the largest calibration deviation in the gallery.

B Optimizer and stopping-rule measurements

Setup. Every number in this appendix comes from one configuration, so that only the optimizer variables change. The system is the design-amortized GBM. The model is the big one (2.3M parameters). The score is FID_σ against the exact conjugate reference on a fixed six-bucket evaluation set. The design sizes are $K = 2, 5, 10, 20, 50, 95$, with 32 observation sets per bucket and two-chain floors of 0.0012–0.0019. Reported values are means over the six buckets. Each training run occupies one H200 GPU. Several comparisons hold the example budget (batch $\times$ steps) fixed on purpose and say so.

Step size at the reference batch. At batch 512 and 9,000 steps (4.6M examples):

Muon step η	3×10^{-4}	10^{-3}	3×10^{-3}	10^{-2}	2×10^{-2}	6×10^{-2}
FID_σ	0.0133	0.0070	0.0065	0.0061	0.0085	diverges

The basin is broad and shallow between 3×10^{-3} and 10^{-2} . The edge beyond it is hard. The package value is 10^{-2} .

Scaling with batch size. We repeated the sweep at batches 64, 2,048 and 8,192, with the example budget fixed within each comparison. The optimum is at 3.5×10^{-3} , 10^{-2} (flat to 2×10^{-2}) and 2×10^{-2} . From batch 64 to 512 the optimum moves by $\times 2.82$, against $\times 2.83$ predicted by $\sqrt{B}$. The fourth-root alternative is excluded, not only disfavored: its prediction costs 0.0079 against 0.0066 at batch 64. Past batch $\sim 2,048$ the optimum stops moving. At 8,192 the square root would extrapolate to 4×10^{-2} , which scores worse than the capped 2×10^{-2} (0.0060 against 0.0052), and twice the extrapolation diverges. Hence the rule of Sec. 4: $\eta(B) = \min(10^{-2}\sqrt{B/512}, 2 \times 10^{-2})$, square root below the critical batch, constant above it.

Transfer across width. At batch 512 and fixed budget, we compare the step 10^{-2} with the hotter 2×10^{-2} across three widths. The results are tiny 0.0089 vs 0.0087, small 0.0066 vs 0.0076, big 0.0061 vs 0.0085. At the hotter step the accuracy is non-monotone in model size (big is worse than small). At the step of the rule the accuracy follows the capacity. “Wider is worse” can therefore be a statement about one number inside the optimizer and not about capacity.

Steps against examples. At a fixed budget of 9.2M examples, batch $512 \times 18,000$ steps reads 0.0051, batch $2,048 \times 4,500$ reads 0.0045, and batch $8,192 \times 1,125$ collapses to 0.0076. At the same batch and step size, twice the steps (2,020) recovers 0.0060. Above a step floor the example budget governs the quality. Below it no example budget helps. The floor is per-system and spans an order of magnitude. The pharmacokinetic validation curve reaches its optimum near 66,000 steps at batch 512. The GBM curve shows no interior optimum up to 36,000. This fact motivates validated stopping in place of per-system budgets.

Validated stopping. The resolution of the monitor must be evaluated at the sample count of the monitor itself. The GBM validation set resolves 0.0013 at 4,000 draws but only 0.0044 at 1,000. We measured both failure modes of the naive keep-the-best rule before we adopted the margin. A monitor coarser than its model ships the deepest noise dip: 0.0053 on test against 0.0040 for the final weights. A resolving monitor still ships 0.0052. Under a cosine schedule the mid-run checkpoints are systematically inflated, and an argmin over the whole curve mistakes that for a reversal. The margin rule accepts the best checkpoint only when it improves on the final weights by more than the resolution of the monitor. It passes both cases. A genuine late reversal clears it by an order of magnitude. On pharmacokinetics the validation curve turns from 0.1066 at step 66,222 to 0.1321 at 72,072, which is twenty floors. The selected checkpoint recovers the difference.

C Reference posteriors

GBM conjugate posterior. With observations at indices $i_0 = 0 < i_1 < \dots$ of the fine grid, the log-returns $r_k = \log(S_{i_{k+1}}/S_{i_k})$ over gaps τ_k are independent $\mathcal{N}((\mu - \sigma^2/2)\tau_k, \sigma^2\tau_k)$. In the coordinates $(b, \sigma^2) = (\mu - \sigma^2/2, \sigma^2)$ the model is a Gaussian location–scale family with sufficient statistics $T = \sum_k \tau_k$, $R_1 = \sum_k r_k$ and $SS_w = \sum_k r_k^2/\tau_k - R_1^2/T$. Draws from the standard conjugate posterior ($\sigma^2 = SS_w/\chi_{n-1}^2$, $b \mid \sigma^2 \sim \mathcal{N}(R_1/T, \sigma^2/T)$) are transformed to (μ, σ) and importance-resampled to the uniform box prior with the Jacobian weight. We validated the sampler against exact-likelihood MCMC on the same designs. Over 24 comparisons, the worst posterior-mean discrepancy is 0.099 sd and the width ratios are 0.91–1.03.

Adaptive Metropolis. Random-walk Metropolis on the box, with proposal $x' = x + sC\xi$, $\xi \sim \mathcal{N}(0, I)$. During burn-in ($\max(500, n/2)$ iterations) the scalar s adapts toward a 0.25 acceptance rate in blocks of 50 with a decaying Robbins–Monro gain. Once enough burn-in states are collected, the Cholesky factor of the empirical covariance, rescaled by $2.38/\sqrt{d}$, replaces the shape C (Haario et al., 2001). Adaptation is frozen for the retained phase. Out-of-box proposals are rejected, which samples the box-truncated target exactly because the proposal is symmetric. The likelihoods are as follows. OU uses per-gap Euler–Maruyama Gaussian transitions, which match the generative process, plus the stationary density of X_0 . GBM uses the exact log-return density. Merton uses the Poisson-mixture transition density, truncated at the numerically negligible jump count. Pharmacokinetics uses log-normal residuals around the Bateman curve. Fisher–KPP uses a Gaussian observation model around the deterministic PDE solve. Every chain used as a reference is paired with a second chain from a well-separated start. The between-chain posterior-mean discrepancies are reported next to the results they certify.

Nested sampling, SMC, and particle-filter likelihoods. FitzHugh–Nagumo and Hodgkin–Huxley references come from `dynesty` (static nested sampling; the prior transform is the uniform box; `nlive=1000`). The admission partner runs `nlive=2000` from an independent seed, and the pair must agree below the two-chain threshold of Sec. 5.1. Stochastic Lotka–Volterra and Heston use adaptive tempered SMC (Del Moral et al., 2006). The inverse-temperature step is chosen at each stage so that the effective sample size stays at half the population. Systematic resampling and random-walk rejuvenation at the current temperature follow. Admission is by population doubling (N against $2N$ from independent seeds). Their likelihoods are unbiased Monte-Carlo estimates. Lotka–Volterra uses a Durham–Gallant bridge sampler (Durham and Gallant, 2002). Heston uses a bootstrap particle filter that runs the clamped-Euler update of the simulator itself (Sec. 5.1). The particle counts keep the spread of the estimator at fixed parameters to a fraction of a nat. Double well and polynomial drift use the same bridge estimator inside the pseudo-marginal variant (Andrieu et al., 2010) of the adaptive Metropolis sampler above.

D Correspondence with package options

in this report	in the package
point tokens	<code>embed="wpoint"</code>
pair tokens + set conditioning	<code>embed="wfilm"</code>
warped values/increments (fixed designs)	<code>embed="wbasis" / "wdiff"</code>
attention phases from physical time	<code>rope="time"</code>
value coordinate of the token (raw or log)	<code>DesignProblem.value_coord</code> , required
fresh designs each optimizer step	<code>fit(retokenize=...)</code> via <code>make_retokenizer()</code>
Muon with the batch step law	<code>fit(optimizer="muon"); muon_lr=None</code> → the measured law
validated stopping with margin	<code>fit(keep_best=True, monitor_floor=...)</code>
sampler steps, scheme and time grid	<code>sample(n=, n_steps=, solver=, t_grid=)</code> , all required
evaluation-set resolution at n draws	<code>evaluation.floor_at</code>
variable-design problem wrapper	<code>amortix.designs.DesignProblem</code>
benchmark systems	<code>amortix.problems (DESIGN_Z00)</code>
calibration screen	<code>sbc_design</code>
reference machinery	<code>amortix.mcmc, .smc, .bridgelik, .hestonlik</code> + likelihood factories

`FlowPosterior(problem)` selects the embedding and the attention phases from the problem class. The settings that decide the result of a sampling call have no default values. The number of draws, the number of ODE steps, the scheme and the time grid are named in every call. The coordinate of the observed value is declared by the problem class. The non-default architecture options stay in the package as reproducible controls.

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